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NONVERBAL BEHAVIOR AND NONVERBAL COMMUNICATION [1]

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In this paper an attempt is made (*a*) to articulate some of the suppositions implicit in different approaches used to study nonverbal behaviors as communication, (*b*) to reemphasize a conceptual distinction between nonverbal behaviors which can be considered as communications and other nonverbal behaviors, (*c*) to reformulate some issues in the study of such behaviors, (*d*) to establish definitions and criteria which can make possible some empirical investigations of nonverbal communication, and (*e*) to indicate how these definitions and criteria can be applied to the study of hand and arm movements as communicative gestures. It is hoped that the availability of the conceptualization, along with some means for operationalizing the concepts, will make it possible to initiate systematic empirical investigations of nonverbal communication in general.

In recent years there has been a marked resurgence of interest in behaviors variously called "body language," "expressive movement," and "nonverbal communication." The inclusion of the terms "language," "expression," and "communication" in these phrases suggests an act of transmission of information from one person to another; that is, the terms imply a subject who is actively making his experience public to some other person. The terms, as used, also suggest that there is a set of nonverbal behaviors which have shared meanings [4] in the same way that there is a set of verbal behaviors which have

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[4] It is important to note that we will use the terms "code," "referent," and "meaning" without implying a particular bias for one or another theory of meaning or theory to account for the ways in which the code component and its referent are related. While the way we use the terms may seem to imply a preference for a referential theory of meaning, we are not committed to this view (see Alston, 1964, for his categorization of theories of meaning). As far as we can determine, the issues raised here about code usage seem to apply equally to all theories of language. Fundamentally, we are concerned about the problems involved in trying to "discover" a "language" rather than to explain how it is possible to use language.

shared meanings. It would be reasonable, then, to expect that investigations conducted under the rubric of nonverbal communication would be concerned with the transmitting subject (encoder) and with the behaviors (code) by which he makes his experience public to another (decoder). Instead, a review of the extensive literature in this area indicates that most investigations seem to be concerned with the significance which some observer can attribute to a particular behavior; that is, the emphasis seems to be primarily on decoding. Even given a decoding approach, there is little consensus about any set of behaviors which can be considered to serve communication functions, and the literature consists, for the most part, of a fragmented and unsystematic array of reports, with almost any conceivable behavior considered by one or another investigator to have some communicative significance. Thus, such diverse movements as body or head positions, lint picking, foot kicking, scratching, gross postural shifts, and hand and arm movements such as palm up during a verbalization have been considered equally cogent and relevant for an investigation of communication.

Whatever the conceptual approach (e.g., transactional, psychoanalytic) or the concerns of the investigators (e.g., behaviors of individuals or of groups), most if not all investigators seem to take a decoding perspective. Investigators who share this perspective also appear to share an assumption that if the observer can make an inference about an individual from his behavior, then the behavior can be considered to be a communication. Unfortunately, this kind of implicit assumption seems to fuse the notion of sign with the notion of communication.[5] For us, "sign" implies only an observer making an inference from, or assigning some significance to, an event or behavior, while "communication" implies (*a*) a socially shared signal system, that is, a code, (*b*) an encoder who makes something public via that code, and (*c*) a decoder who responds systematically to that code.[6]

When investigators have not clearly distinguished sign from communication, the statement, "I take x to be a sign of y" becomes transformed into, and considered equivalent to, the statement, "He communicates y via x." The logical difficulties in this transformation can easily be seen when it is applied to inanimate events. For example, the fact that one may take dark cumulus clouds to be a sign of impending rain does not mean that the clouds communicate, *via a code* which includes darkness and cumulusness, that it will soon rain. Nevertheless, investigators have frequently transformed statements such as "I take behavior x to be a sign that he is angry" into statements like "He communicates his anger via behavior x." It is clear that while any and all behaviors of an individual or group may be used by an observer as a basis for making inferences (i.e., any behavior may be taken as a sign of something else), communication implies something about the activity of the behaving one and the specific significance of the act.[7]

Movements are Expressive

This kind of confusion among signs, inferences, and communications can better be

[5] The issues discussed in this paper are relevant primarily to those investigators, whatever their concern or their philosophical or theoretical orientation, whose definitions distinguish, either implicitly or explicitly, communication or language behaviors from other behaviors. For those investigators who make no such distinctions, for example, those who treat *all* behaviors as signs, some of the issues discussed in this paper may not be relevant (see Footnote 7 also).

[6] By "code" we will mean only a shared set of behaviors which have shared referents, as in Morse code. We will use the term "system" in this context, at least, to include the code and the principles for usage of code elements. For us, as for others, "language" additionally implies some rules governing organization and combination of code elements (i.e., grammar, syntax).

[7] While we will go to considerable length to distinguish sign and communication behavior, this distinction is important only if the concern of the investigator is with communication per se. If the concern is with understanding the behavior of a particular individual, signs and communications may be equally valuable data sources, and the distinction between these two sets is not very relevant. If, however, the concern is with communication—that is, a system independent of particular individuals or particular referents—then these distinctions are important.

exemplified by showing two ways in which the phrase "movements are expressive" can be interpreted. One interpretation takes the phrase to mean: "In principle, under specified conditions, everything a person does, every action or movement he makes, can be used by a trained, knowledgeable, and sensitive observer as a basis (either from the behaviors themselves or from inferences based on these behaviors) for categorizing (or inferring about) the individual—his state, thoughts, feelings, personality, or cultural patterns." In this context, the phrase "everything a person does . . . can be used as a basis . . . for inferring about . . ." can be abstracted to the statement, "If behavior x occurs, an observer may infer a concomitant state y, or an antecedent z, or both, as *correlates* of the behavior x."

A second interpretation of the phrase "movements are expressive" takes "expressive" to mean that a behavior is a manifestation of something which is itself not observable. The apparent reasoning in this interpretation is that since an observer can infer something (e.g., a genotypical characteristic, psychodynamic conflict, or group membership) about the speaker from a movement—much as he can from a spoken word—then what has become evident to the observer must have been *made* manifest by the actor. The movement is then considered to have the characteristics of a communication; that is, something is being made public by the doer. This interpretation of "movements are expressive" then involves a shift from an observer making an inference, given certain behavioral events, to a subject who is manifesting something, that is, communicating via these behaviors. In this interpretation, there is yet a further transformation. What is being made evident is known by the observer who makes the inference, while the doer apparently neither knows what is being made manifest (his assumed communication) nor does he make any inference about his movement. For this reason, the phrase "movements are expressive" is further transformed to read: "A person, in doing x, is revealing y covertly." In this form, the behavior is seen as a manifestation of something covert which is being involuntarily communicated or revealed.[8]

For investigators who use this second interpretation of the phrase, if something is covert and being revealed, it seems reasonable to infer that what is being revealed must be unacceptable or it would be expressed in a more overt form. Further, these investigators reason that if something is covert, then its behavioral manifestation must be in a form which should not be recognizable by the actor, that is, symbolic in the psychoanalytic sense. In final form, the phrase "movements are expressive," as interpreted from this perspective, seems to mean: "A person's behavior x communicates in some symbolic form his unacceptable thoughts, affect, impulses, etc."

The first interpretation of "movements are expressive" was concerned with an observer who makes inferences about traits of the subject, about concurrent cognitive or affective experiences of the subject, and/or about antecedents of the behavior. In contrast, the transformations of the second interpretation have subtly shifted the focus to a subject who intentionally or unintentionally symbolizes unacceptable content by means of these behaviors. In these transformations there is also a shift from an observer who is knowledgeable about a set of categories, uses them reliably, and has a set of correlated events with which he associates the categories in his observation, to an observer who has been specially trained to interpret symbolic forms of expression within the context of some particular theory (e.g., psychoanalytic).

While the transformation patterns described above may seem like word games, an examination of the literature on body language, expressive movements, and nonverbal communication would indicate that all too often an observer's inference seems to be interpreted as a subject's communication, either explicitly by the investigator

[8] It is all too tempting to infer unconscious communication from movements, since the subject often seems to be neither as aware of his movements nor as able to reconstruct them as he is aware of and able to reconstruct his spoken words. The issues of intentionality and awareness are discussed later.

himself or by some ambiguity in the conceptualization which makes it possible for a less sophisticated reader to make the transformation. The shift from an inference-making observer to a communicating subject takes different forms, depending on the behaviors being investigated (e.g., idiosyncratic vs. socially shared) or the theoretical assumptions of the investigator (e.g., psychoanalytic theory vs. social learning or role theory). We will attempt to exemplify the transformation from sign to communication as it occurs in the context of a number of representative approaches to the study of nonverbal communication by referring to investigators who have been trying to understand the individual and to investigators concerned with group behavior.

In selecting our exemplars, we have tried whenever possible to select the most influential investigators, many of whom have also been concerned with the very transformation issue with which we are concerned. However, we will attempt to demonstrate that despite the concern for distinguishing communicative from noncommunicative nonverbal behavior, the decoding perspective which has been the primary focus of investigations in this area can be seen as a main source of confusion in the field in general. Simultaneously we hope to explore within each example additional issues which we feel are important if we are to arrive at a conceptual framework which will permit some systematic investigations of nonverbal communication from an encoding perspective. In any case, none of our criticisms, either implicit or explicit, should be taken as minimizing the importance of the contributions of the investigators whose works we are analyzing, for without their pioneering work there would be nothing to extend.

Decoding Perspectives with Focus on the Individual

The writings of many of the psychoanalytically oriented investigators clearly show the way in which the transformation from movement as sign to movement as communication could occur.[9] Given the assumptions and the context within which psychoanalytically oriented investigators have traditionally worked, it is seductively easy to make such a transformation. The patient's task in the psychoanalytically oriented situation is to make his past and present experience public via verbal language; the analyst's task is to understand or "make sense of" the patient's report. To understand, in this tradition, is to seek indexes of content and experiences which are not being verbalized Thus, with verbal material it is the latent rather than the manifest content which is taken to be most critical. This concern with latent content goes with the assumptions that everything which is available to consciousness should be communicable through consensually shared verbal forms, and that anything which is not available to consciousness (i.e., anything being defended against) will be evident only in a form in which the patient himself could not recognize the content—that is, the material is in symbolic form.

From our viewpoint, the latent content may be seen as an inference drawn by the therapist based on the patient's verbalizations. Although we agree that the manifest content is a communication, when verbalizations are interpreted within a psychoanalytically oriented context, they seem to be used as behaviors, with the behaviors treated as signs of some other particular state, affect, or experience. What we would take to be an inference seems to be taken by these investigators to be some latent content. It would seem, however, that in this psychoanalytically oriented view, both manifest and latent content are often treated as equivalent in terms of communications. For both types of content, many psychoanalytically oriented observers seem to see themselves as decoding a communication encoded by and emanating from the patient. Indeed, to talk about latent content at all seems to presuppose that those meanings are "in" the utterance and, by extension, "in" the speaker, rather than that those meanings are the listener's inferences. The transformations thus can move subtly from "I can infer

[9] There are clearly a number of psychoanalytically oriented investigators who have been explicit in separating their inferences from the patient's communications (e.g., Mahl, 1968).

x from his utterance" to "his utterance communicates x" to "he communicates x symbolically and thus unconsciously." To the extent that such transformations occur for verbal material for any psychoanalytically oriented investigators, the same transformations could occur even more easily when dealing with nonverbal behaviors, especially if the nonverbal behaviors are taken as prima facie manifestations of some latent content that is by definition unable to be verbalized. The patient can then be seen as using nonverbal behaviors unconsciously to communicate that which he cannot or will not communicate verbally. Deutsch (1959) made explicit this kind of assumption: "Postural behavior is already a stage of transformed instinctual drives, a kind of acting out in a rudimentary way of expression that could not reach the verbal level [p. 129]." Feldman (1959), Reik (1954), and Robinson (1965) seem to have taken positions very similar to this one.[10]

There are a number of interrelated problems for understanding nonverbal communication as a system which arise if nonverbal behavior is considered to be unconscious encoding of experience. First, if the unconscious encoding is considered to be idiosyncratic (i.e., if each person encodes his experience in a different way), then there seems to be little if any possiblity of deriving a socially shared system for encoding experience in movements. If a consensually shared system of encoding experience is taken to be one of the criteria of a communication system (as in verbal language), idiosyncratic unconscious encoding would not meet this criterion. From a research concern, the view that each individual encodes his experience in some idiosyncratic behavior could only yield a series of case studies showing how a particularly insightful and experienced observer has cracked a particular idiosyncratic code. Such intensive case studies are common in many psychoanalytically oriented contributions to the literature on nonverbal communication. However, there appear to be no principles determining which behaviors are selected for study or interpretation by the investigator, and there seems to be little overlap of research efforts on the significance of particular movements. It is unclear how any number of case studies could increase our knowledge of nonverbal communication as a system.

Second, if unconscious encoding of experience is assumed to take consensual rather than idiosyncratic forms, then this assumption would at the least be more consistent with most views about communication. However, this view requires some consideration of the status of the decoders. It could be assumed that there is a select group of decoders, that is, decoders (e.g., psychoanalysts) who know a code that others do not know. This assumption would result in an unusual distribution of encoders and decoders. In most communication groups, there seem to be more decoders than encoders of the communication system. For example, children decode verbal language forms which they do not use in encoding, and beginning students of a foreign language can often decode more than they can encode. Thus, the notion of a communication system which contains all members of the group as encoders, but only a few as decoders, seems contrary to what we find about en-

[10] Certain procedures occurring in psychoanalytically oriented psychotherapy seem almost guaranteed to elicit material which is consistent with psychoanalytic assumptions about manifest and latent content, and which thus precludes questioning of the assumption of unconscious communication. Therapists frequently note particular behaviors of their patients, either from direct observation or from the patient's report of past behavior. To understand the patient better, the therapist will then ask him to talk about his experience concurrent with that behavior. For example, a patient may be directed to "tell me how you felt (what you were thinking, etc.) when you did x (some particular behavior)." Most patients will respond to this directive with some verbal communication about some experience. It is tempting then to assume that the nonverbal behavior was a communication of what was later communicated in consensual verbal forms. It is even easier to maintain this assumption if the therapist has made an inference about the patient's experience from the behavior, and this inference is supported by the patient's subsequent verbalizations. If verbalization does not occur, the therapist may still infer resistance or repression, thus supporting his assumption that the behavior was a nonverbal communication.

coding and decoding within other communication systems (e.g., verbal language). Additionally, the assumption of consensual unconscious encoding requires some empirical documentation of the posited significance of a movement. Further, if every movement were taken to be an unconscious consensual encoding of experience, any research effort would leave most movements unstudied, and no finite number of experimental studies would justify logically the generalization that all movements are universally encoded communications. If only some movements are to be taken to be encodings, it is difficult to know the basis for determining which movements are relevant for study.

It is possible, of course, to assume that all behaviors are consensually unconsciously decoded as well as consensually and unconsciously encoded, with only a few members of the communication group able to decode consciously what others encode and decode unconsciously. If this view of nonverbal behavior as communication is taken, it is incumbent upon the conscious decoders to specify at the very least the explicit criteria they use for determining the consensual referent of particular behaviors. To our knowledge, no attempt at such specification of criteria has been made.

In sum, the kind of approach to communication exemplified by many psychoanalytically oriented investigators is so broad that any and all behaviors are considered to be communications. It is difficult to see in this approach any basis for developing a systematic study of nonverbal behaviors as communication given (*a*) some of the assertions in the theory and the procedures of those who use the theory in this way, and (*b*) the fact that assigning of a behavior to the category of communication seems to be based solely on the observer's ability to make an inference from the behavior. If no distinction is made between signs and communications, it is unclear how such an approach could contribute anything to the study of communication as a special system or as a special set of behaviors. In fact, there seems to be no logical basis for excluding from communication instances any occurrence about which the observer might make an inference. This failure to distinguish sign from communication, and inference making from decoding, would yield a fragmented and unsystematic research literature, whose contribution to the study of communication as a special system is uncertain. In our view, it would seem that an attempt must be made to distinguish sign from communication conceptually before communication can be investigated empirically in any systematic way.

There is a group of investigators (e.g., Dittmann, 1962; Duncan, 1969; Sainsbury, 1955) who do not seem to be concerned about making the distinction between movements as signs and movements as communications. In contrast to the psychoanalytically oriented investigators discussed above who seem to treat all behaviors as communications, the latter investigators appear to treat all behaviors as signs. For example, Sainsbury correlated the occurrence of movement in any part of the body with indexes of affect (e.g., verbalization, galvanic skin response); Dittmann correlated the occurrence of head, arm, and foot movements with verbal indexes of affect; and Duncan (personal communication, 1969) correlated movements from various parts of the body with each other and with verbalizations. However, since these investigators also use participants in a therapeutic interview as subjects, the temptation is ever present for a reader to view the observed behaviors as communications of affect or other momentary states, much as these transformations are made by some psychoanalytically oriented investigators. Whatever the view of these investigators about the relevance or importance of making a sign–communication distinction, subsequent workers in the field appear to have taken these investigations as studies of communication, as evidenced by the inclusion of such studies in the literature on nonverbal communication.[11]

[11] The studies cited above, which use correlation techniques and which imply that movements are expressive of other states, are similar to the earlier work of Allport and Vernon (1933) and Wolff (1935). These earlier investigators correlated movements and movement styles with other

Incidentally, while a correlational research strategy seems appropriate for the investigation of movements as signs or indexes of enduring or momentary states, this strategy seems less appropriate for a study of movements as communications. A communication system, by our definition at least, requires that (*a*) individual units of behavior be specified, (*b*) each of these behavior units have specific significance, and (*c*) for the most part, the significance of each unit be different from the significance of other units. In the correlative studies cited above, all movements within any one part of the body have been treated as interchangeable, with each movement thus having no special significance (e.g., Dittmann, 1962; S. Duncan, personal communication, 1969), or any movement of one part of the body has been treated as equivalent to any movement of any other part of the body (e.g., Sainsbury, 1955). It is difficult, then, to see how this kind of approach can be heuristic for the investigation of communication as a system, were anyone concerned about these data for the study of communication. The results of the all-behaviors-as-signs approach, like those of the all-behaviors-as-communications approach, have been either (*a*) a series of case studies in which particular movements are found to be correlated with particular indexes for a particular person in a particular context (e.g., Dittmann, 1962; S. Duncan, personal communication, 1969), or (*b*) a general conclusion such as movements in general express affect in general (e.g., Sainsbury, 1955). The case studies approach appears to yield findings which make it difficult to arrive at any statement of shared or concensual use of movements; the general statement, by its very all inclusiveness and lack of specificity, makes further research almost unnecessary. In sum, a movement-as-signs approach, however painstaking and time consuming, would not seem to lead to understanding of nonverbal communication, if communication is taken to be a socially shared system for the encoding of experience and its decoding by another member of the group.

Some investigators (e.g., Krout, 1935a, 1935b; Mahl, 1968) have distinguished conceptually between movements as signs and movements as communications. Krout distinguishes conventional gestures, which he defines as socially patterned and shared forms of behavior, from autistic gestures, which he defines as movements which are unique for a particular individual in a particular context. Mahl similarly distinguishes communicative gestures, which he defines as movements which everyone would agree are substitutable for a verbal utterance,[12] from autistic gestures, defined as movements which are not substitutable for a verbal utterance. In his conceptualizations, Mahl has included these autistic gestures in the class of informative movements. Despite these explicit attempts to distinguish between movements as signs and movements as communications, the use of a decoding approach in studies of gestures makes it difficult to maintain this distinction. As we have suggested, there does not seem to be any way of knowing from an observer's perspective whether one is decoding a communication encoded by the individual or whether one is drawing an inference from his behavior.

We hold that these difficulties are inherent in a decoding perspective, even for investigators who attempt to distinguish signs and communication, and can best be exemplified in some work of Krout (1935a, 1935b) and of Mahl (1968) with autistic gestures.

behaviors (e.g., handwriting) which in turn were viewed as indexes of a relatively enduring state (personality, character). The writings of these earlier investigators seem to have focused on individual differences and on the consistency of the behavior of an individual over time, and seem to have been less concerned with nonverbal communication. Nevertheless, these investigators' description of their subject matter as expressive movements seems to have prompted the inclusion of their work in the literature on nonverbal communication by some authors, and has influenced the research strategy of later investigators.

[12] At first glance, the terms "conventional gestures" and "communicative gestures" may appear to refer to the same class of events. However, Krout's category also includes a great many socially patterned movements (e.g., bread buttering, car driving) which do not appear to us to meet Mahl's criterion of substitutability for a verbal utterance.

Krout (1935a, 1935b), for example, states explicitly that "autistic gestures have no definite meaning, either for the subject or for the individual to whom the subject is responding [1935a, p. 401]." This statement seems to imply that autistic gestures are behaviors, neither encoded nor decoded. Krout further hypothesizes that these behaviors are explicit responses to internal stimulation, and he demonstrates through a variety of experiments that (*a*) a particular idiosyncratic behavior is related to a specifiable external stimulus (word), (*b*) the external stimulus is related to other external stimuli (words), and (*c*) from the exploration of the word association configurations an observer may infer the existence of an emotional conflict or complex within the subject. However, when Krout finds evidence of an emotional conflict/complex and comes to the conclusion that an autistic gesture is a disguised expression of other nonexpressible behaviors or of nonexpressible words (i.e., those behaviors or words whose overt manifestation has been inhibited), the reader is likely to associate these kinds of statements with unconscious expression of repressed material. The collapse of the sign–communication distinction for the reader is made even more probable when Krout (1939) states,

> Theoretically, autistic gestures occur because they give us temporary relief from tension and conflict. To gesture means to think frankly along certain lines, and to *express* our thoughts with impunity. By means of self-directed gestures, we can say things about ourselves which we are loath to regard as attempts at personal advertising [p. 172].

Although Krout at times states that his goal is interpretation of autistic gestures, when he later says that to gesture "means" or "expresses," he seems to imply that he has decoded the subject's self-communicative acts. In this way, the transformation from sign to communication seems to have occurred; if Krout himself has not made it, the unwary reader is likely to. Thus, Krout starts with a conceptual distinction between conventional gestures and autistic gestures which appears to be based on a distinction between behaviors as communications (socially shared patterns of encoding and decoding which serve to make experience public) and behaviors as signs (behaviors serving as a basis for inference). However, in coming to the conclusion that autistic gestures can be seen as possible substitutes for verbal utterances to oneself or others, he could be seen as attributing the inferences he makes about the behaviors to an encoding process on the part of the subject, blurring, at least for the reader, the very distinction between signs and communication which he had previously attempted to introduce.

Mahl (1968) clearly warns that an

> Interpersonal orientation to psychiatry . . . places one in the danger of viewing everything the patient does that is *informative* to the therapist or observer as though it were also *communicative* in motivation [p. 334].

However, his own use of terms and his analyses of behaviors make it difficult for the reader to maintain the distinction he so carefully posits between informative and communicative movements. In his own work with autistic movements, Mahl finds that inferences he has made on the basis of the behaviors are later validated by some verbal communication from the subject. He then seems to conclude that some autistic gestures can be seen as precursors of later verbalizations; that is, given the appropriate evidence, an autistic gesture can be interpreted as involving "spontaneous nonverbal anticipation of verbalization [p. 323]." Since Mahl's definition of communicative gestures specifies that the behavior can be substituted for a verbal utterance, and since his analysis of autistic behavior may well lead the reader to the conclusion that these behaviors are alternative forms of expression for that which is later expressed verbally, it would seem that the distinction between communicative and autistic gestures can be maintained by the reader only with great difficulty.

Whether or not there is a collapsing of the sign–communication distinction for Krout or for Mahl, the readers of these investigations have collapsed the distinction, as evidenced by the inclusion of these studies in the literature of nonverbal communication.

We hold that insofar as communication at the very least involves socially shared behavior patterns, the study of autistic behaviors seems nonrelevant to the study of nonverbal communication. To us, the very inclusion of studies of autistic movements in the literature on nonverbal communication is evidence that our concerns are not unwarranted. We note, however, that Mahl and Krout in their work on autistic gestures, and indeed the other approaches which have been discussed thus far, have been concerned with understanding a particular individual. With a focus on understanding an individual, anything the individual does or says can be used as a basis for inference about the person, and any distinction between sign and communication is an unnecessary nicety. When communication is the focus, the distinction is mandatory in our view (see Footnote 7).

While some investigators have been concerned with nonverbal communication primarily to make more sense of particular individuals, Ekman (1964, 1965a, 1965b) and his associates (e.g., Ekman & Friesen, 1968)[13] seem to have been more explicitly concerned with communication per se. Ekman has made a noteworthy attempt to focus explicitly and systematically on communication. Nevertheless, the early formulations offered by him (at least up to and including 1968) appear to include many of the conceptual and methodological problems discussed thus far. Like others who employ a sign approach to movements, Ekman and his associates employed a definition of communication which seems synonymous with the way in which we have used the concept of sign. Specifically, Ekman and Friesen (1968) state that

> the term *communicate* refers to the fact that observers are able reliably to decode information from viewing a sample of nonverbal behavior. There is no implication that the person enacting the nonverbal behavior intended to communicate nor any assumption that the communication is necessarily accurate [p. 186].

In this statement, a behavior is said to be communicative if a number of observers make the same inference on the basis of the behavior. Although behaviors such as gait as a sign of age, or hip sway as the basis for inference about sex, could logically be studied as communication within this definition, these investigators focus on making inferences about affect from films or photographs of subjects in interview situations.

The fact that such behaviors as gait and hip sway are not included in their behavioral samples suggests that there are some implicit assumptions about communication which may not be included in the more explicit definition. Ekman and his associates have sampled behaviors only in the special context of a psychiatric interview. In this particular interpersonal context (which many investigators of nonverbal behavior have used as a data source), verbal communication is taking place. It seems that movements which have shared meanings are more likely to occur in a verbal communication context than in the many other contexts (e.g., a ball game) which might have been sampled.[14] For Ekman and his associates, as well as for other investigators, the very choice of a verbal exchange as the context for sampling behavior has obscured some of the difficulties associated with defining communication on the basis of observer consensus. At the same time, it makes their particular definition and approach appear to be reasonable for the study of nonverbal communication.

Another problem highlighted in some of this earlier work of Ekman and his associates, but evident also in the work of many other investigators of nonverbal communication, is the assumption, implicit if not explicit, that nonverbal behavior in general is communication about affect, or somehow related to affect (e.g., defense against it). Assuming that nonverbal behaviors have significance only for communicating about

[13] The most recent work of Ekman and Friesen seems to involve a radical shift in perspective. We discuss this work in a later section of this paper.

[14] It is logically possible to hold that there is an inverse relationship of verbal to nonverbal communication instances (as in the hydraulic model), or that these two sets of behaviors are uncorrelated. However, the definition we will propose would lead us to expect most instances of nonverbal communication behavior to occur in social verbal communication contexts, rather than in social situations per se.

affect appears to have limited to one domain the possible range of decoding or inference making by the observer. For example, in this earlier work Ekman and his associates used either a two-category system for inferring pleasant and unpleasant affective states (Ekman, 1965a, 1965b) or an adjective checklist of words such as "nervous," "fearful," "angry" (Ekman & Friesen, 1968). It is not clear on what grounds—logical, empirical, or otherwise—the significance possibilities of nonverbal communication have been limited to affect significance, while verbal communications are recognized as having the possibility of a variety of domains of significance (e.g., informational, indicative). This implicit constraint seems to have unnecessarily hindered the exploration of the possible communicative significance of nonverbal forms of behavior, even within a decoding definition of communication.

Another methodological point in these earlier works of Ekman, as well as in the work of others, deserves mention. In some earlier studies (e.g., Ekman, 1965a, 1965b) the visual stimuli which judges rated were selected from interviews by a time-sampling procedure. Such a procedure is consistent only with an approach which defines communication solely in terms of observer consensus (i.e., if any behavior on which a consensus can be reached is taken to be a communication, then any and all behaviors are equally relevant for study). From our viewpoint, however, this procedure is likely to sample both communicative and noncommunicative movements. It seems hardly possible, using this sample procedure and rating system, to differentiate (*a*) observers' *decodings* of communications encoded by the interviewee, and (*b*) observers' *inferences* from the interviewee's behavior.

One further difficulty in an approach such as that of Ekman and his associates involves the same kind of transformation from observers' inferences to subject's communication already noted for other investigators. In this earlier work, Ekman (1965b) explicitly disavows the possibility of studying encoding in communication, as in the following statement:

> "What the sender actually experiences, intentionally or nonintentionally, is not only almost impossible to determine, but is in no way necessarily equivalent to the indicative or communicative value of his nonverbal act [p. 393]."

However, there are several indications that Ekman and his associates, like most others, seem to have assumed implicitly that the subject has encoded what the observers inferred. For example, the very use of the word "decoding" implies that there is or has been encoding. Similarly, the use of such phrases as "the subject conveys" or "the subject transmits" also implies active encoding on the part of the subject. Further, the failure to obtain observer consensus is explained by invoking encoder variables, as in the following: "there might be limitations to nonverbal communication, either for this patient or in general [Ekman & Friesen, 1968, p. 169]." None of these terms (i.e., decoding) or statements (i.e., limitation of patient competence in nonverbal communication) appears consistent with the disavowal of concern with encoding on the part of the behaving subject. However, there is some indication that Ekman and Friesen are now more explicitly concerned with encoding in their study of nonverbal communication. In a recent paper (Ekman & Friesen, 1969), a category system for movements was offered which depended almost entirely on encoder variables as criteria for categorization. However, it is not clear how this latter approach will be made coordinate with their former work, nor how it will influence their ongoing research strategy. We recognize, with Ekman and Friesen, the difficulties involved in investigating encoding; we show how we propose to deal with these difficulties in a later section of this paper.

Decoding Perspectives with Focus on Shared Behaviors

In contrast to the investigators who focus on the behavior of individuals, another group of investigators included in the literature on nonverbal communication focuses on behaviors common to a group.[15] The behaviors of interest to the latter investigators are at least known to be shared; they are thus, from our standpoint, more likely to be acceptable as communicative instances than are behaviors specific to particular individuals. Nevertheless, we hold that it is difficult to determine which, if any, of these socially shared behavior patterns qualify as communications by our definition. Again, from our perspective, investigations of socially shared behavior, like the investigations of an individual's behaviors, have proceeded from the standpoint of the observer or decoder. Accordingly, the subtle transformation from observer's inference to subject's communication is as easy to make in investigating shared behaviors as it is in investigating individuals' behaviors. There would at first seem to be some advantage, for communications research, to focusing on socially shared behaviors. This advantage disappears if the research approach makes it difficult or impossible to explicate a basis for identifying a socially shared communication code, in contrast to learned social behaviors or rule-following behaviors.

While many investigators of shared behaviors seem to make the transformation from observer's inference to subject's communication, they differ somewhat from each other in terms of the number and kinds of behaviors they include in their study of communication. Some investigators seem to include in their investigations only those behaviors which occur in interpersonal contexts, while still others focus on behaviors co-occurring with verbal communication in an interpersonal context, with each successive limitation on the range of behaviors making the transformation seem less obvious to a reader. We attempt (*a*) to explicate how the transformation operates for some different investigators of these social behaviors, and (*b*) to consider some of the problems attendant upon each approach. In addition, we hope to make some further distinctions which we hold are necessary if we are to arrive at a definition of communication which will be consistent for both verbal and nonverbal behaviors, and if we are to be able to initiate systematic investigations of communication.

The work of Goffman (1959) is a good example of the way the transformation from sign to communication can occur (either for the investigator or for the reader) when the investigator seems to treat all socially patterned behavior as communication instances. Goffman points out that a number of behaviors can be reliably correlated with a person's position or role in the social system. He gives "insignia of office or rank; clothing, sex, age, and racial characteristics; size and looks; posture; speech patterns; facial expression; bodily gestures; and the like [Goffman, 1959, p. 24]," as examples of signs which can be used by an observer as a basis for inference about the person's membership in a particular group. Even type and placement of living room furniture can be included among the events which Goffman considers relevant for his concerns. When Goffman treats these behaviors and events as signs of group membership, there are no difficulties in accepting his insightful comments on the social scene. However, he often calls these same behaviors or events "sign vehicles for expressing" and "communications." The interchangeable use of the terms "sign," "sign vehicle," and "communication" may well lead a reader to believe that all sign behaviors are transmissions of information by the subject, rather than bases for inference by an observer. The transformation from sign to communication appears plausible in that (*a*) the correlates of the behaviors cited are often known by most members of the group, (*b*) members of the group sometimes report having taken these signs into account when

[15] For this group of investigators, as well as for the others we discuss, we are concerned only with the possible confusion of the sign–communication issue in their writing. We recognize that the problem may sometimes be one for the reader rather than for the investigators themselves. Whether our criticisms are valid or not, at no point do we question the value of the insightful observations made by those investigators of the behavior of individuals or of particular groups in particular settings.

interacting with another, (*c*) the behaviors whose correlates are known can be and sometimes are used by one person to manipulate another into attributing to the user these correlates of the behavior, and (*d*) Goffman uses the metaphor of the theater to demonstrate his use of signs.

There are several difficulties which can be exemplified in this approach. First, while the correlates of some signs are generally known by members of the group[16] (e.g., the wearing of a mink coat is correlated with wealth), the correlates of other, equally frequently exhibited signs are not known except by a few expert observers. For example, some hair-grooming behaviors appear to be correlated with gender; males typically groom hair by running their fingers through it, while females are more likely to pat their hair into place. If the correlates of a behavior are unknown by the emitters of the behavior, it is difficult to see how the behavior (e.g., hair grooming) could be used to communicate the correlate (e.g., masculinity or femininity in this case) even though it is a sign for the expert observer. Second, even if the correlate of the behavior is known and the behavior is sometimes used, it would seem less than logical to conclude that the behavior is always a self-conscious enactment.

There is a more basic concern about an approach which involves sign behavior, but which implies enactment. For a sign to be predictive, a correlation between the sign event and the other attribute which is to be designated must already be evident; that is, the co-occurrence must be known prior to any individual's using the sign event. For example, it must first be noted that the "Jet Set" wear certain clothes before anyone new can use the wearing of such clothes to be identified as a "Jet Setter." We cannot distinguish, from the clothes wearing itself, whether the wearer is part of the original group or is using his knowledge of the correlation. Further, to hold that a behavior learned by a group is learned so that the group membership of the individual will be identifiable seems to us to distort what is meant by learned behavior sequences. We call someone a "Southerner" when he speaks with a particular pattern of intonation; the Southerner does not learn to speak with that particular pattern of intonation so that he will be identified as a "Southerner." It would seem strange to say that he enacts or uses a Southern accent rather than simply that he has learned to speak that way. Similarly, one group of girls may learn to wear particular kinds of clothes. When an observer notes some characteristic of this clothing which distinguishes it from the clothing of other girls, we can construct a classification in terms of his observation. This classification is introduced by the observer, not by the clothes wearers, even though the observer could now take the clothing as a predictor of subgroup membership. We would hold, then, that instances in which a sign is used for manipulation purposes are infrequent. Although some signs are sometimes used for manipulation or deception, it does not seem reasonable to treat all signs as if they were always used or enacted.

There is a second group of investigators (e.g., Birdwhistell, 1970) who also study socially patterned behaviors, particularly movements. Unlike some investigators who seem to fuse sign and communication, Birdwhistell has in some of his early writing explicitly distinguished expression from communication in terms of a sign–communication distinction.[17] From our perspective, one of the most important issues raised by this group of investigators is that they explictly hold that nonverbal communication could most fruitfully be considered and analyzed in the same way as are verbal communications. Given this commitment and focus, these investigators seem to select movements of all kinds and attempt to determine the basic units which can be considered the building blocks of all movements, much as the phoneme is considered the equivalent

[16] See later discussion under section "An Alternative Approach to Nonverbal Communication." The same logic which is offered to determine code use can be applied to the question of "Knowing" or "Known" by a respondent.

[17] This distinction is sometimes blurred in Birdwhistell's anecdotal reports of social behaviors.

building block of words. In a sense, these investigators view their work as being analogous to the work of structural linguists who deal with verbal-vocal material. While investigators of verbal language and these investigators of movement share a common methodology, we can note an important difference between them. The structural linguists apply their analysis to verbal events which have already been accepted and designated as meaningful events. For the structural kinesicist, on the other hand, there is not, at least to our knowledge, a set of movements which have been designated as meaningful to which a structural analysis can be applied. If a structural analysis is to be applied to movements, then, it must either be applied to all movements, irrespective of their communicative value, or to those movements which the investigators have taken to be meaningful, in which case it seems likely that the selection of movements would be based on what the investigator can make sense of and therefore assume to be decoded. In any case, the criteria for selecting particular movements to analyze have not been spelled out by these investigators. Instead, their writings imply that the use of a structural analytic methodology will, in and of itself, lead to the discovery of which movements are meaningful and what the meaning of those movements might be.

We have no doubt that a structural analysis of events already designated as part of a communication system may be extremely helpful in explicating the principles for representing, encoding, and decoding within the system. However, it is unclear to us how a communication system can be specified via the use of a structural analysis. For example, while we agree that it is possible to analyze every vocal utterance into some combination of phonemes, we may still ask how it is possible to specify which combinations of phonemes are to be classified as morphemes in any particular communication system. Similarly, while we agree that it is possible to analyze any and all movements into some combination of kines, we are still in doubt as to the method or criteria by which it would be possible to conclude that the combination ABC is a kinemorph, and thus part of a particular movement communication system, while the kine combination ABD is not a kinemorph within that communication system. In sum, although we can understand how to proceed with a structural analysis after a communication system has been designated, we do not know how to discover a communication system by an analysis of the smallest common units of a modality.

Thus, if there is no a priori specification of the set of movements to be analyzed structurally, then Birdwhistell (1970) and others are left with the impossible task of analyzing every movement. Even were this task possible, it is still not clear how an analysis of structure could lead to specification of meaning of the occurrences unless it is assumed that structure and meaning are correlated in some as yet unspecified way. Taking verbal behaviors as the paradigm of communication, this latter assumption appears to be less than tenable.[18]

Birdwhistell (1970) seems to recognize the difficulty inherent in trying to determine the meaning of a movement solely on the basis of a structural analysis of the movements, since he notes that similar movements may serve different functions and different movements may serve similar functions. However, his solution to this dilemma seems to be a belief that the meaning of a movement will be discovered by a careful analysis of the total context(s) in which the movement occurs. Again, it is no clearer how either a structural analysis or a detailed specification of the context will be a basis for deriving meaning—unless the meaning of the context is already known or posited. Incidentally, Birdwhistell appears to compound somewhat the difficulties associated with this approach when he holds that the meaning of the movement will be known only when the *total* context is known. This

[18] In verbal communication systems, the relationship between symbols and referents appears to be quite arbitrary; that is, no consistent principles for relating symbol and referent are apparent. It seems unlikely that the relationship between symbol and referent in nonverbal communication systems would be any less arbitrary.

latter task seems logically impossible. It is not clear how an investigator can know that he knows the total context. We hold with Wiener and Mehrabian (1968) that since we do not know whether there is anything we do not know, we cannot be certain we know the total. Empirically it would seem that this search for the total context would be endless, for it will always be possible to include something else in the total, as Birdwhistell seems to do when he now includes olfactory components and micromomentary muscle changes as part of his concern with the total context.

Somewhat tangentially, it seems to have become fashionable for some investigators of nonverbal communication to use instruments which allow much more detailed inspection of behavior than would be possible for the average observer of the behavior (e.g., Haggard & Isaacs, 1966). The use of such devices might be helpful for detailed description of the observed events, or even for drawing hypotheses of correlations between particular aspects of the behaviors and other ongoing events. It is difficult, however, to see how minute movements discovered with special equipment could be included as part of a communication system, if these movements could not even be observable by participants in ordinary interactions.

In sum, Birdwhistell's (1970) observations about socially shared patterns of movement have been most insightful, and his delineation of the particular muscles involved in specific constellations (e.g., smile) for particular subgroups has been informative. However, it is not evident how this type of structural approach will lead to the specification of the movements which are to be considered as communications rather than signs, nor how the meaning of any movement will be determined.

There are a number of investigators (e.g., Scheflen, 1964a, 1964b, 1965, 1967; Watzlawick, Beavin, & Jackson, 1967) whom we categorize as taking as nonverbal communication any behavior occurring in an interpersonal setting. While Scheflen sometimes asserts a distinction between communicative and informative movements, at other times he seems to be espousing a definition similar to Watzlawick et al., who explicitly define communication as "any behavior occurring in the presence of another." Investigators who hold this view seem to justify the use of the context (interpersonal setting) as defining communication behaviors, by trying to show how the behaviors which do occur in this setting can be understood by an observer. In showing how they make sense of the behaviors, they, like many others, seem to be making the transformation and fusing the notions of sign and communication which from our viewpoint require separation. For example, Scheflen (1965) shows how a particular behavior sequence can be understood as a within-family communication event: A mother leans forward to speak to a therapist; the father begins to tap his foot; the grandmother and the daughter (flanking mother) cross legs toward each other; the mother stops talking to the therapist and leans back. In an attempt to make sense of this recurring sequence, Scheflen supplies the following meanings of the behaviors: When the mother leans forward she is flirting with the therapist; when the father taps his foot he is signaling to grandmother and daughter to limit the mother's flirtatiousness; when the grandmother and the daughter cross their legs toward each other they are telling the mother to stop flirting; when the mother leans back she is saying that she understands their message.

From our standpoint, if these behaviors are communications they should be part of a code shared either by a cultural group or at the very least by members of this family. Since Scheflen (1965) does not report the occurrence of this behavior sequence in other families in therapy, it would seem to be, at best, a code which is used only by this family. However, other than the interpretation offered, no evidence is presented that code usage is involved. We hold that for the behaviors to be considered a code in our sense requires the possibility of (*a*) interchangeable use of code elements by different members of the family, and (*b*) occurrence of the same code elements in contexts other than the one reported. No evidence is presented that, for example, the grandmother taps her foot to tell the daughter to limit the mother's flirtatiousness, or that a leg cross in another context has the meaning of stop it. It is, therefore, difficult to accept the view that this particular behavior sequence does involve a code, even one shared only by this particular family. Further, without evidence that the emitters of the behaviors are encoding meanings via a shared code, it would seem that the meanings in the behaviors are inferences of the observer rather than that the behaviors are the subjects' communications.

In contrast to Scheflen's (1965) account, let us see how the same behavior sequence can be accounted for in terms of learned responses—an analysis in terms of cue–response learned sequences—rather than in terms of a special communication code. Mother leans forward when talking to a male; whenever mother talks to another male, father becomes anxious or restless; when he is anxious or restless, he taps his foot; when father taps his foot, grandmother and daughter recognize that he is upset (i.e., they recognize that in the past this sign behavior has been correlated with instances of upset); they are uncomfortable when father is upset; when they are uncomfortable they fidget, crossing legs, etc.; mother notes the moving around and takes the fidgeting and other signs of restlessness as discontent (or upset or whatever else it has correlated with in the past); she stops what she is doing, which makes it possible for the others to enter the interaction.

Since this and similar behavior sequences can be accounted for without reference to communication, it would seem unnecessary to account for these kinds of behaviors as communication. In all fairness to investigators such as Scheflen, the behaviors selected for analysis are those which ostensibly occur with some frequency and have some predictability of sequence in particular interpersonal contexts.

If we examine some of the apparent assumptions underlying the definition and selection of instances for study, we may begin to understand the general acceptance of this type of approach. Verbal behaviors can elicit feedback from others, the patterns of verbal exchanges are somewhat predictable, and verbal behaviors involve the communication of meanings. From the perspective of this group of investigators, the nonverbal behaviors in interpersonal contexts which they select also seem to elicit responses from others, and these nonverbal behavior sequences also seem somewhat predictable; the nonverbal behaviors are then apparently assumed to have the same status as verbal communications. If this logic is reduced to syllogistic form, it would read approximately: All A (the nonverbal behaviors in interpersonal contexts) are B (predictable, feedback eliciting, etc.); all C (communications) are B (predictable, feedback eliciting, etc.). The conclusion that all A are C is clearly untenable. The apparent logic in an approach that includes all behaviors in interpersonal contexts as communications suggests that the justification of this kind of definition is on tenuous grounds. We recognize that definitions are arbitrary—subject only to agreement and not to proof. However, if all behaviors which could occur in the presence of another person are to be considered to be communications, what then is *not* communication? If communication is defined in such a way that *all* behaviors are to be included in the category, there is nothing to be gained from labeling this all-inclusive category as "communication" rather than simply as "behaviors" in an interpersonal context. Incidentally, it is not always clear whether investigators who hold this definition mean that all behaviors which occur in the context are communications, or that only those behaviors of which they can make sense, which reoccur, and which elicit predictable responses from the other people in the context will be included.

There is a group of investigators who define nonverbal communication as those behaviors which co-occur with the verbalizations in an interpersonal or face-to-face interaction. The issues raised for investigators who define the nonverbal communication instances by the interpersonal context are equally applicable here, even though this group has further limited the set of nonverbal behaviors labeled communication to only those behaviors that accompany the verbal communication. Given that the instances of concern involve explicit communication (i.e., something is being made public via verbal content), the probability is increased that at least some of the movements occurring may also be communications. The problem, however, is that this group of investigators, like the previous groups, gives no independent operations for deciding which if any of the nonverbal behaviors involve code, encoding, and decoding.

The work of Efron (1941) seems to exemplify best the approach of this group. Starting with the observation that the hand and arm movements of two groups from different language backgrounds also differed in terms of size, shape, and tempo, Efron began to investigate the movements themselves. By limiting himself to movement co-occurring with verbalizations, Efron was able to posit the referents of some of the movements, pointing out that these movements could refer to objects, actions, ideas, relations between objects or ideas, or to the flow of ideas. To the extent that Efron did not check his inferences about these movements as code elements, standing for referents, his inferences have the same limitations as those of other investigators who have studied communication solely from an observer's perspective. Nevertheless, to the extent that the behaviors observed were exhibited by several members of a given group, that these behaviors seemed to occur only in the context of a verbal exchange, and that they seemed to be related to the ongoing verbalizations, such behaviors seem more likely to be possible instances of communication from our perspective than the behaviors selected for study by most other investigators. They appear even more promising as communication instances, since several independent observers have noted similar behavior patterns in different groups and at different times (e.g., Ekman & Friesen, 1969; Saitz & Cervenka, 1962; Wiener, 1968). Despite this promise, however, systematic work is still required to make it possible to conclude that this behavior set is a code and part of a communication set. Later in this paper we outline one possible approach toward this end.

In sum, the investigators of socially patterned behaviors have described a number of behaviors and behavior sequences which could well be communications by our definition. They have not, however, clearly discriminated learned behaviors from behaviors which have referents, or habitual response sequences from sequences peculiar to communicative exchanges. They fail to make these discriminations primarily because they, like the investigators of behaviors of the individual, seem to focus on observer's responses to or inferences about the behaviors, rather than considering the subject's use of the behavior to refer to a momentary state or experience. The investigators of socially patterned behaviors have generally justified calling the observed behaviors "communications" by pointing out that the behaviors in question show many of the characteristics of verbal communication: regularity of occurrence, patterns of elements or organization, or response-eliciting characteristics. These investigators have not yet demonstrated that the behaviors in question share with verbal communication the function of referring to something else. The end result is that in a sense these investigators have defined communication in such a way that almost anything could be considered a communication. If communication is to be considered as a subset of behaviors in general and of socially patterned, interactive behaviors in particular, some further definition is required which will make possible the discrimination of behaviors which serve a function similar to that served by verbal communication—namely, the making public of experience via a set of socially shared signals—from those that do not. As we have noted, this discrimination will require some consideration and investigation of encoding, for we hold that the distinction between inference making and decoding cannot be made without some independent evidence of encoding.

An Encoding Approach

In a context which has belabored the point that there have been all too few attempts to investigate communication from an encoding perspective, it is encouraging to note that Ekman and Friesen (1969) seem to be moving toward making distinctions similar to the ones discussed here. Starting with and elaborating somewhat on Mahl's (1968) earlier distinction between informative and communicative behavior, Ekman and Friesen now define communicative nonverbal behavior as

> those acts which are clearly and consciously intended by the sender to transmit a specifiable message to the receiver [in contrast to] informative nonverbal behavior [which is defined as] those acts which have some decoded meaning in that such acts elicit similar interpretations in some set of observers [p. 55].

Within these definitions, there is a clear attempt to separate activities on the part of the behaving subject from activities on the part of the observer; a necessary first step from our perspective in making a sign–communication distinction. Unfortunately, however, while Ekman and Friesen's definition clarifies the sign–communication distinction in some respects, the further explication of these terms introduces ambiguities and confusion in other respects. For example, in these very definitions, Ekman and Friesen do not distinguish the activity of decoding from the activity of inference making, using the terms "decoded meaning" and the making of "interpretations" interchangeably. We note again that any behavior can be a source of information on the basis of which observers may make inferences, but that only some behaviors are decoded. Insofar as Ekman and Friesen do not include a notion of code in their concern about informative events and inference making, the concomitant use of a term such as "decoding," which would seem to imply a code, can only lead to confusion for a reader.

Similarly, in their definition of communication, Ekman and Friesen (1969) seem to imply that encoding on the part of the subject is a prerequisite for considering a behavior to be communicative. We would agree with this prerequisite. However, while the use of the term "encoding" would again seem to imply use of a code, Ekman and Friesen seem to assign a rather different and somewhat ambiguous meaning to this term, as in the following statement: "The act has an encoded meaning in terms of the regularity of its occurrence with those stimulus events which precede, accompany or typically follow it, or with consistently associated ideation [p. 54]." Part of this statement seems to indicate that Ekman and Friesen consider encoding to involve not the use of a code, as the term implies, but regularities in the association of a set of conditions with a particular emitted behavior. If Ekman and Friesen mean by "regularity" that the conditions of occurrence of the behavior can in principle be specified, then all behaviors could be considered regular (and thus, in the particular conceptual framework, encoded). If the term "regularity" involves a notion of frequency of correlation of conditions with occurrence of behaviors (i.e., a notion of predictability), then every simple conditioned response (e.g., eye blinks prior to onset of air puff) must be considered to be encoded. We would prefer to reserve the term "encoding" to refer to regularity in behavior involving the use of a code—the code being specifiable independent of particular instances of occurrence.

Although Ekman and Friesen use the term "coding," their explication of this term appears to preclude the possibility of identifying a code. Ekman and Friesen offer three ways in which a behavior can be related to a referent (i.e., coded), namely, (*a*) arbitrarily, in which the relationship of the behavior to the referent is one of consensus; (*b*) iconically, in which the behavior resembles some attribute or characteristic of the referent; and (*c*) intrinsically, in which the behavior has no referent other than itself. To the extent that Ekman and Friesen seem to imply that coded behavior is used to signify some referent, it is difficult to see why they have included, in coding, behaviors which have no referents other than themselves. Again, in our framework, a code is taken to be a set of behaviors which have referents other than themselves; with this concept, it makes no sense to include, as coding, behaviors which do not have such referents. To imply that every behavior is coded in some way, as Ekman and Friesen seem to do, dilutes the importance of both the concept of code and the concept of coding, and leads once again to the possibility of including any and all behaviors as part of a communication system.

Without an explicit concept of code to aid in the sign–communication distinction, Ekman and Friesen seem to have fallen back on the conscious intentions of the subject as the criterion for considering a behavior to be communicative. Even though Ekman and Friesen recognize that the concept of intention is fraught with empirical problems, they imply there is no apparent alternative.

Since it is not apparent to us how the intention of the communicator can be known by an observer, this concern seems at best irrelevant. If intentionality is defined by asking the communicator to indicate his intentions, then only behaviors which he says he intended as communication can be designated as communicative. In this instance, then, there is no problem. If, however, the observer maintains that the speaker may be unaware of his intentions (i.e., unconscious intention), mistaken about his intentions, or dissimulating, then there is no basis in the behaviors themselves for distinguishing between communicative (intended) and noncommunicative (nonintended) behaviors. In a larger context, it is clear that the distinctions between the truth, a lie, and a mistake are decided on grounds other than the behavioral event or the person's report. To cite a classic example, was Joan of Arc telling the truth about her visions, was she mistaken about what she saw, or was she lying? As is obvious in this instance, the observer must assert his own assumptions in drawing his inferences about the truth value of an event. Until Ekman and Friesen offer some satisfactory epistemological basis for determining the intentionality of the communicator, we see no reasonable solution to the problems inherent in a definition of communication which includes an undefined term, intention. By restricting their use of intention to include only conscious intentions, Ekman and Friesen avoid some of the problems noted above, and state in a footnote that they hope to arrive at some empirical criteria for distinguishing a consciously intended behavior from other behaviors. However, to the best of our knowledge, the specific criteria have not yet been explicated.

Despite ambiguities in the use of the terms coding, encoding, and decoding, and the lack of an explicit concept of code, Ekman and Friesen have begun to specify sets of behaviors which, in their frequency of occurrence, are at least candidates to be considered as part of a communication system. That these latter sets of behaviors have also been delineated by Efron (1941) and have been included in a system of categorization we will offer would seem to indicate that the behaviors may be independent of particular encoders and decoders, thus meeting a second criterion for consideration as part of a communication system.

Incidentally, Ekman and Friesen (1969) have noted (as has Scheflen, 1964a) a class of behaviors which they call regulators. As described by these investigators, this set of behaviors seems to serve the unique function of regulating the communication interaction itself. The set of behaviors which Ekman and Friesen call regulators seems to include the kinds of eye contact behaviors studied by Exline, Gray, and Schuette (1965), what Hall (1963) has called proxemics, or what Scheflen (1964a) has called general interaction patterns. If these behaviors do in fact regulate verbal communication, their occurrence can serve as the empirical basis for holding that encoding and decoding are occurring nonverbally. We hope in a later section of this paper to explicate more fully this possibility. However, in the absence of an explicit specification, by Ekman and Friesen or anyone else, of the set of behaviors which serve to regulate a verbal or nonverbal communication, it is all too easy to include as regulators all behaviors of a person to which another person responds or might respond—as some investigators have done.

An Alternative Approach to Nonverbal Communication

Having considered the problems associated with both decoding approaches to communication and an encoding approach with a focus on intention, let us now propose an alternative conceptual model which we hold will lead to a somewhat different research strategy for studying nonverbal communication than has apparently been used thus far. Like other investigators of communication, we are faced with the problems of (*a*) developing a definition of communication, (*b*) operationalizing the concepts in the definition, (*c*) selecting behaviors to study, and (*d*) explicating a research strategy to explore the implications of the proposed approach.

We have thus far in the paper rather loosely defined communication as the making public of experience via a shared code for encoding and decoding. The major concepts at issue in this definition are code and encoding, for only after these concepts are more fully explicated can the concepts of decoding and communication be understood within our perspective. We start with a definition of code which is essentially similar to a dictionary definition of code as a set of arbitrary components which have referents other than themselves, for example, words, Morse code (see Footnotes 4 and 6). From this definition, it follows that a dictionary of such elements and their referents could in principle be compiled. Such a dictionary would be expected to change over time, however, in the same way that a dictionary of verbal forms changes. Further, while we could reasonably expect some behaviors to have more than one possible referent (as in verbal language some words have more than one referent), we assume that the number of referents for any given behavior would be finite and in principle specifiable, just as the several meanings of a given word are finite and specifiable. We further assume that the specific referent of a particular behavior in a particular context may be determined by its relationship to the rest of the communication matrix, again just as the particular meaning of a word in a given instance can almost always be specified on the basis of its relationship to other code components in the communication matrix. For example, the word "rage" has different meanings in the sentence "The man flew into a rage" and in the sentence "The miniskirt is the rage." The specific meaning of the word rage in these sentences depends on its relation (syntactic-semantic) to other components. These assumptions about symbols and referents are clearly at odds with a view which holds that the "meaning" of any behavior can be known only through knowledge of the total context in which it occurs, or with a notion of idiosyncratic code.

Since we assume that the relationship between the code component and its referent is most often arbitrary, that is, any behavior A, B, or C can be used to refer to any experience X, Y, or Z, as long as the participants agree that a given behavior (e.g., A) stands for or refers to a particular experience (e.g., X), we would not expect any nonverbal communicative behaviors to have universal significance, as some investigators apparently imply (e.g., Ekman & Friesen, 1971). Rather, we would expect cross-cultural differences and even subgroup differences in (*a*) the experience to be made public, (*b*) the nonverbal behaviors used to make experience public, and (*c*) the relationship between experience and the particular nonverbal behaviors. To find invariance of behavioral form and referent across cultures would to us be prima facie evidence that the behavior is not part of a coding system. We could agree that some forms of behavior (e.g., facial variations) might be more likely than others to be used in making public some sets of experience (e.g., mood state). However, a behavior whose significance is found to be invariant across cultures (i.e., observers attribute the same significance to the behavior in all cultures) would be difficult to include as a code component, which for us involves an assumption of an arbitrary relationship between symbol and referent. Rather, we would subsume such behaviors under the rubric of stimulus-specific behavior sequences—even if the behaviors occur under different conditions in different cultures (as suggested by Tomkins, 1962, 1963; and investigated by Ekman & Friesen, 1971; Ekman, Sorenson, & Friesen, 1969; Izard, 1968, 1969).

We assume further that the relationship between behavior A and experience X, for a particular communication group, must at the start of the relationship between A and X (or in developing any formal code) have been made with some awareness, although the experience of self-conscious use of A to stand for X may dissipate for the group as well as for the individual with increasing association between A and X. That is, we view this shift to nonawareness as being no different than the equivalent change in awareness in driving a car. As experienced drivers, we are typically not self-conscious or aware of the movements of hands, arms, and feet which are involved in steering in a particular direction—rather, we focus on the direction to be steered and the appropriate movements occur almost automatically. That these behaviors were once in awareness and learned, however, can be testified to by anyone who remembers learning to drive. We hold that the self-consciousness involved in communicative behaviors is evident only for relatively inexperienced communicators. Anyone who has become proficient in the use of a second language will recall the difficulty he once had in finding the proper word to fit a particular referent—an example of self-consciousness in encoding which diminishes with increasing experience.

As we have pointed out, other investigators have also held either that nonverbal behavior constitutes a code which is decipherable, or that behaviors involved in communication must be socially shared. We would not, then, expect much disagreement with a definition of communication which includes the use of a shared code in which the behaviors constituting the code refer to experiences other than themselves. The difficulty and disagreement, then, if there is any, would be about the operations employed to infer the instance of code and the related concepts of encoding and decoding. In sum, if a set of behaviors will be considered to constitute a code, it must be demonstrated both that the behaviors have referents and that the referents of the behaviors are known and used by a group; that is, used by at least two persons who emit (encode) the behavior to stand for the agreed-on referent and take (decode) the emitted behavior of the other to stand for the agreed-on referent.

To the extent that for us encoding means using a set of code components which make experience public, we must specify the operations or behaviors which we will take as indicators of code usage. Without such specification of behaviors, it is all too easy to assert code usage whenever we see code components. Such an assertion might be acceptable if the code components were independently specified and agreed on. However, since nonverbal code components have not been specified and agreed on, investigation of encoding via nonverbal forms requires some set of operations which make it reasonable to infer code usage.

The method we employ to operationalize code usage seems analogous to the procedures suggested by Hofstadter (1941) under his rubric of "objective teleology." [19] The following extended quotation from Hofstadter seems to explicate best the logic of his approach.

> . . . from the methodological standpoint it seems advisable to develop a way of identifying purposive action without linking it to any specific explanatory hypothesis concerning the internal structure of the purposive agent . . . . As in any problem of identification the best procedure consists in observing and analyzing cases of action which are definitely recognized or labelled as teleological, in order to make explicit the common traits which they exhibit. . . .
>
> There is, then, no initial difficulty in locating objective teleological processes in the rough. The problem is, what common traits do these actions exhibit? In particular, where in these actions do we find objectively purposeful character? And the answer is, we never find an objective purpose by itself, but always in association with a certain "sensitivity to conditions" and a fund of "operative techniques" possessed by the actor. To seek for objective purpose alone, without reference to these two factors, is to embark upon an impossible quest. A purposeful action is directed to its end always in a concrete set of circumstances and along paths of connection between antecedents and consequences. Differences between purposive actions rest not only upon differences of ends, but also upon the range and depth of the circumstances or conditions which enter effectively as well as upon the scope of the connections of antecedents and consequences actually operative. This becomes especially clear when considered from the viewpoint of prediction. In the case in which a teleological action is not as yet complete, if I merely assume that the actor is pursuing a cer-

[19] Though our method is similar to Hofstadter's, our concerns are not with the philosophical problem of teleology. We are indebted to Leonard Cirillo for pointing out to us the similarity between our approach and Hofstadter's.

tain end I can not from this predict the methods and procedures it will employ nor the occasions and matters upon which it will employ them. I need to assume that it is "sensitive" to certain circumstances in specific ways; and I need also to assume that it has a fund of operative techniques, of abilities to connect antecedents and consequences, i.e., that it will respond to a specific condition to which it is sensitive in a specific way. It is possible that many different actors should have similar ends and yet behave in radically diverse ways. The differences of behavior are not predictable, and do not follow, from the similarity of ends, but only from the ends plus the differences in sensitivity to conditions and in the funds of operative techniques available to the actors. But since, in predicting, I need to be able to predict differentially, it follows that my assumption of end alone does not provide me with an adequate empirical hypothesis. On the other hand, if to the assumed end I enjoin a specific sensitivity to conditions and a fund of operative techniques, I do get an empirical hypothesis capable of differential prediction and therefore susceptible of confirmation or disconfirmation. . . . To discover operative techniques in teleological process in general requires a number of contexts of activity in which the subject of investigation appears and in which ends and sensitivities to conditions vary. Regularities in the choice of similar means for similar ends under similar conditions, different means for similar ends or similar means for different ends under different conditions, different means for different ends under similar conditions—all these indicate technological principles in operation, operative techniques. . . .

The unitary attribute of the teleological actor is not the possession of end alone, or sensitivity alone, or technique alone, but all three in inseparable combination. However, it is also true that although they can not be separated in the unitary attribute they may nevertheless be analyzed out independently by the use of a plurality of acts of the same agent. It is possible to discover the fund of generally operative sensitivity and technique possessed by the subject of investigation by observing its behavior in varying contexts, where there are convincing reasons to believe that ends vary, and segregating what is constant in sensitivity or in technique. By analogous methods one can study the general character of its ends by varying the contexts so as to call for variations in sensitivity and technique. It is thus possible to discover constancies in any one of the three factors with variations in the others.

It must be emphasized that ascription of a unitary objective teleological attribute, consisting of end, sensitivity, and technique, to an object is in no way an explanation of its behavior. It is simply a shorthand way of describing that behavior. The behavior exhibits a pattern of directedness, passing through certain specific circumstances within a context, and proceeding by certain connecting paths. The directedness is a directedness of the paths through the contextual circumstances. Directedness without paths and circumstances is nothing. Paths in no direction and connecting no circumstances are nothing. Circumstances not located on paths leading in a direction are nothing. "End" is a name for the directedness, "sensitivity" for the circumstances, "technique" for the connecting paths, when taken in relation to the agent. The problem of explaining the action arises only when one asks whether a correlation can be made between these elements of the actions and structural-dynamic properties of the actor. The existence of the teleological process is sufficiently attested when a hypothesis about end, sensitivity, and technique is confirmed [pp. 32–35].[20]

Hofstadter appears to be arguing that an event can be considered goal directed (*a*) if some observable set of criteria can be stated (in this instance the sensitivity of the doer, the conditions in which the act occurs, the means, and some end state), and (*b*) if modifying any one of these leads to a predicted change in the others. The underlying principle involved in his approach to investigating behaviors or events which have been traditionally defined in subjective terms, then, seems to be that of deriving a set of operational events which are posited as being related, and then checking outcome predictions made on the basis of this posited interrelationship when one of the relationships among the designated events is changed. If the predictions are confirmed in a variety of instances in which the relationships among criterial events are manipulated, then Hofstadter holds that goal-directed behaviors have been investigated both empirically and objectively.

As we have noted above, our method for operationalizing encoding or code usage involves a strategy similar to that proposed by Hofstadter. That is, code usage will be posited on the basis of a set of interrelated predicted findings when manipulation of the experiences to be made public, of the forms available to the addressor, or of the responsiveness of the addressor to communication conditions results in the predicted effect on the emitted behaviors as a function

[20] Reprinted with permission from an article by Albert Hofstadter published in the *Journal of Philosophy*, 1941, Vol. 37. Copyrighted by the *Journal of Philosophy*, 1941.

of such manipulations.[21] Specifically, this method involves five interrelated steps. First, we start with behaviors which are consensually taken to be an instance of code usage—namely, verbal language behaviors.[22] Second, we identify a set of behaviors associated with verbal language behaviors. The occurrence of this set of behaviors is taken to be independent evidence for, and consistent with, the assumption that encoding is taking place in verbal behaviors. That is, we will identify behaviors from which we can conclude that the encoder's verbal behavior is systematically modulated as there is a change in the experience to be made public, in the conditions of the communication, or in the response patterns of the decoder. Third, we demonstrate that in the same way that predictable introductions take place within verbal language (i.e., variations in words occur as a function of systematic manipulation of the experience to be made public, of the conditions of the communication, or of the decoder's response patterns), predictable nonverbal behaviors will be introduced when the verbal possibilities are constrained. Fourth, we show that the introduction of these nonverbal behaviors does not result in changes in those decoder behaviors which we take to be indicators of decoding or understanding in verbal exchanges. Fifth, we argue that if these nonverbal behaviors are introduced predictably, and if their introduction does not result in a significant change in decoding indicators, then the nonverbal forms introduced can be considered to be substituted for verbal forms, and thus can themselves be considered as components of a nonverbal code whose emission involves code usage.[23]

Four classes of behavioral events can be identified whose occurrence coincides with our assumption that verbal language involves code and code usage. Two of these classes, search and correction, appear to be indicators of a code, that is, a standard against which emitters check their verbalizations. The other two classes, consistent with an assumption of code usage, are (*a*) regulators, one set emitted by the addressee to indicate decoding and another set emitted by the addressor to check on whether decoding is occurring, and (*b*) message modulations, which are responses to variations in the event to be made public, to the conditions of communication, and to the regulators or other behaviors of the addressee. These four classes of behaviors appear in both verbal and nonverbal forms during verbal interactions. In describing these classes of events, the four classes will be presented as if they were independent occurrences, but for the most part two or more usually seem to co-occur, either sequentially or concurrently.

Search typically involves a pause greater than the stops required for paralinguistic designation of word, phrase, or sentence; such a pause may also occur at points other than paralinguistic junctures in the flow of verbal communications. The pause is usually accompanied by the speaker looking away (usually up) while maintaining a constant tonal pitch (the pitch neither drops, as at the end of a declarative statement, nor rises, as at the end of a question). At the end of the pause, the speaker reinstitutes the flow of verbal communication. Verbal forms of search may include such statements in the pause as, "I can't think of the right word," or "What's that word that means the same

[21] As we use the terms code usage and encoding, they do not imply goal-directed behaviors, nor do they imply an objective or subjective teleological explanation. At this moment we hold only that if verbal language exchanges are taken to be an exemplar of code usage and encoding behavior, then we can specify a set of operations which will make it possible to hold that particular nonverbal behaviors are code components and part of a communication system in the same way words are. This approach we propose is also analogous to that offered by Cronbach and Meehl (1955) in their discussion of construct validity.

[22] As we have noted earlier (see Footnotes 4 and 6), for us language includes a code, rules for the emission of code components, and rules governing organization of code components.

[23] We recognize the similarity of our substitution criterion to Mahl's (1968) definition of communication. The difference between the two approaches is a function of how substitutability is determined from a decoder versus an encoder perspective.

as—?" We have not noted a significant number of instances of search for nonverbal components, except under special circumstances such as charades or sign language.

Correction is usually noted when a speaker changes an element or component in his ongoing flow of speech. It is most evident in a verbal exchange when an emitter says, "I meant to say ambiguous, not ambivalent," or when an addressee says something like "You mean ravenous, not ravishing, don't you?" When the emitter makes a correction, it is often preceded by search; correction by the addressee typically occurs immediately or as soon as the speaker pauses (i.e., at the end of the thought). Examples of addressee correction of nonverbal forms in communication sometimes occur in response to tonal variations, as when a mother says to her child, "Don't talk to me in that tone of voice," or by an emitter when he says "I didn't mean to say it that way," and repeats the message, changing the tonal variations. As noted above for search, correction of other nonverbal components (e.g., gesture) is also rare, again except in special situations such as charades or sign language. It may well be that the infrequent occurrences of search and correction for nonverbal forms is a function of the relatively small number of components in these nonverbal codes.

Regulators are behaviors which we hold are taken by addressees and emitters as signs that encoding or decoding are occurring, although we do not include these behaviors in the code itself. The set of behaviors we call regulators is more restricted than the set implied by some investigators (e.g., Scheflen, 1964a), in that our set includes only those behaviors which are emitted by most encoders and decoders of a communication group and in most, if not all, communication situations. The set of regulators which we describe here derives in part from the thought-provoking and intriguing study of particular regulators by other investigators (e.g., Exline et al., 1965; Hall, 1963). Although we try to describe each regulator independently, these behaviors also seem to occur in varying combinations. We have no reason to believe that the behaviors we describe are the only regulators which occur, and the set below is not considered to exhaust the category.

Addressor regulators are behaviors emitted by the speaker which appear (*a*) to serve as cues for the addressee that encoding is occurring and continuing, (*b*) to check on whether listening and decoding are occurring, and (*c*) to indicate when the addressee is to speak. The following behaviors have been identified:

A. Eye movement. The addressor typically looks at the addressee at the end of completed thoughts during the flow of speech, with or without a short pause. If the addressee makes eye contact and/or nods at those times, the emitter seems to take this as evidence that the addressee is listening and decoding, and the emitter continues speaking. If the addressee makes eye contact, but with a blank or quizzical look, the speaker typically rephrases or elaborates his message. If the addressee does not make eye contact at these times, the speaker either repeats his message with increased volume or introduces other cues (e.g., throat clearing) which evoke listening on the part of the addressee, or the emitter ceases speaking altogether. However, if the emitter pauses for longer than usual without initiating eye contact, but moves his eyes to the side or up instead, it seems to be taken by the addressee as an indication that the addressor is thinking or searching and will continue his speaking following the pause.

B. Vocal variations. Two particular kinds of vocal variations, inflection and pause, can be considered as regulators. While some of these variations also have paralinguistic significance, we discuss them only as regulators. An upward tonal inflection (as in a question) or a downward tonal inflection (as in a declarative statement), when accompanied by an extended pause, seems to be taken as requiring some response from the addressee—either speaking or some indication of decoding (see below). However, a pause and extended eye movement to the side or eyes rolled up, even with inflection changes, seem to be taken by the addressee as an indication that the emitter is thinking or searching and will continue to speak; no response is required by the addressee.

Addressee regulators are behaviors emitted by the addressee which seem to be taken by the addressor as cues that decoding, understanding, agreement, or disagreement are occurring.

A. Eye contact. The addressee is expected to maintain relatively constant eye focus on the addressor while the latter is speaking. When the addressor pauses and checks the addressee for eye contact, at least four possible responses by the addressee can occur.
   1. The addressee moves his eyes upward without speaking; this behavior seems to be taken as a cue that the addressee is thinking and will respond subsequently. The original emitter waits.
   2. The addressee maintains a quizzical or blank look without speaking, usually taken as a sign that the addressee has not understood the message, and responded to by the emitter with a message variation.
   3. The addressee maintains eye contact with a slow rhythmic head nod. This behavior seems to be taken by the addressor as evidence of decoding or understanding, and the addressor can continue speaking. This nod may be contrasted with a sharp staccato nod, which seems to indicate agreement in addition to understanding.
   4. The addressee maintains eye contact and smiles. This behavior seems to be taken as a cue that the addressee understands but has nothing to add.

B. Other addressee regulators.
   1. A raised eyebrow or a frown seem to be taken as indicating disagreement or questions; the addressor typically reiterates, corrects, or elaborates his message.
   2. A sudden intake of breath by the addressee, co-occurring with the opening of the mouth as if to speak, and sometimes accompanied by a postural shift forward, seems to indicate that the addressee has something to say. This sequence is followed by an abrupt end to the addressor's speech or, if speech continues, by an increase in pitch and/or volume and/or rate of speech. This entire latter sequence is sometimes accompanied by an upheld palm as a pantomime of "stop," and all of this seems to indicate that the emitter has responded to the interruption cue, but will override it and make his point.

These regulators have been observed in a wide variety of situations and for many different addressors and addressees. They are a most interesting phenomenon in communication and deserve more systematic study in their own right. For our purposes here, we only propose to demonstrate that among other manipulations, the control of regulators will lead to predictable variations in nonverbal behavior as well as in verbal communication.

Message modulation within the context of verbal code usage is taken to mean predictable variations in form or style of utterance, or in the components used. These verbal variations are consistent with the assumption of code usage in that the modifications seem to provide evidence that the speaker is taking into account, in his verbalization, changes in the experience to be made public, the conditions of the communication, and the characteristics and responses of the addressee. For example, a young male in describing his activities with a date on the previous evening will use different components when relating the events in a locker room and when speaking to the same addressee in a restaurant. As another example, upon hearing an addressee's foreign accent, or when speaking to a child, an emitter will speak more slowly and clearly and may also use words that are more common than the words he would use with a native adult addressee. As still a further example, a man who complains to his fellow workers using one set of statements will reword the complaint when speaking to the president of his company.

We have briefly outlined a set of behaviors associated with spoken verbal language which will be taken as independent evidence of code usage. Our approach now requires us to define communication more explicitly and then to (*a*) specify a set of behaviors which are hypothesized to be nonverbal code components; (*b*) demonstrate that search, correction, and regulators co-occur with these behaviors; and (*c*) show that the hypothesized code components will occur predictably; that is, they can be seen as substitutions for verbal code components, given the appropriate conditions for message modulations accompanied by restrictions in the use of the verbal code.

In this context, then, communication is operationally defined as the occurrence of encoding and decoding behaviors, particularly as evidenced by regulators. Whether or not understanding does in fact occur, that is, whether emitted behaviors by the communicators have shared referents, may only be known subsequently (e.g., by noting whether subsequent behaviors by both communicators show a correspondence of referents), or by the use of some other criteria (e.g., consensus judgments of other members of the communication group). When mismatches of referents are found, we call such instances code-usage errors. We hold that correction of such errors or mismatches is part of the ongoing learning and elaboration of the code which is available to all members of a communication group. In sum, communication is defined by the ongoing, concurrent behavior sequences of encoding and decoding. Errors in the use of referents are known by some other set of criteria—correspondence of other behaviors or consensus of members of the communication group (e.g., Hymes, 1964).

Given our strategy of prediction, our third step is to denote a set of behaviors—in this paper we outline hand and arm movements—which are possible code components. There are several criteria that such behaviors must meet if we are to consider them as possible components in a communication code, as opposed to reactions to some immediate stimulus or as opposed to a socially patterned behavior (i.e., patterned behavior learned in the process of acculturation):

A. The behaviors must be emitted by the particular communication group studied. This criterion eliminates idiosyncratic behaviors and meets the requirement of socially shared behaviors.
B. The behaviors must occur in several different contexts. This criterion is likely to eliminate some behaviors which are reactions to a specific set of stimulus conditions.
C. The behaviors must be more likely to occur in verbal contexts than in any or all other contexts. If a behavior (e.g., scratching) can occur in any context—that is, with or without an addressee—it is difficult to accept it as a possible component in a communication code.
D. The behaviors as code components should encompass a relatively short time duration. This criterion serves to focus on ongoing experience rather than on socially prescribed patterns of behavior or on behavior related to personality styles (e.g., the wearing of rings or the handling of teacups).

These four criteria for selecting behaviors to be studied make it possible to eliminate a great number of other behaviors which have been considered in the non-verbal-communication literature. For example, these criteria eliminate from our concern the autistic behaviors studied by Krout (1935a, 1935b) and by Mahl (1968), many of the situation-specific behaviors reported by Scheflen (1965) and by Birdwhistell (1970), and many of the habitual, learned social behaviors presented by Goffman (1959).

Further we have found it helpful to utilize the concept of "channel" in selecting behaviors for study. Channel has been defined (Wiener & Mehrabian, 1968) as any set of behaviors in a communication that has been systematically denoted by an observer, that is considered by that observer to have coding possibilities, and that can be studied (at least in principle) independently of any other co-occurring behaviors. The designation of a set of behaviors as a channel in communication is quite arbitrary and is based only on some division of the communicative events which has been found useful for investigation. In essence, then, the designation of a channel has often been dictated by pragmatic considerations or conventionally accepted distinctions, as when body movements are set apart from vocalizations as constituting different classes of events. There is no implication that these specifications of the behaviors as channels preclude other ways of subdividing the communicative event, as long as the perspective of code usage is maintained. Further, no attempt is being made to reify the concept of channel. In sum, then, a channel is a set of behaviors arbitrarily delineated either on the basis of conventional distinctions or on the basis of some other organizing principle.

Using the concept of channel and the criteria for selecting behaviors to be studied, we have isolated the following movements of the hands and arms (gestures). The gestures, along with their posited significance and functions, are listed below. Again, it should be noted that the categories and/or the particular gestures have been described by one or another of the previous investigators in this field (e.g., Efron 1941; Ekman & Friesen, 1969; Krout, 1935a, 1935b; Saitz & Cervenka, 1962).

A. Pantomimic gestures.

1. *Formal pantomimic gestures* are stylized movements of the arms and hands for which there is a culturally prescribed consensual meaning. Typically, such gestures have an object or event significance and function as nouns do in the verbal channel. Formal pantomimic gestures are almost invariant in meaning despite variations in the context, situation, time, or addressee; this relative invariance in meaning is similar to the relative invariance in meaning of words in the verbal channel. Formal pantomimic gestures can take the place of words; when they co-occur with words, the function of the gesture is that of emphasizing by adding redundancy to the verbal communication. Examples of formal pantomimic gestures are waving goodby, depicting two wavy lines in the air to designate a well-proportioned female, "the finger," and forming a circle with the index finger and thumb, remaining fingers extended to signify "okay" or "perfect."

   The number of gestures which have a culturally prescribed meaning is typically limited within the general population, although some subgroups within a culture may make more extensive use of gestural substitutes for words than other subgroups, and may even develop their own formal pantomimic gestures. For example, in nineteenth-century Russia, the sophisticated balletomane knew that a gesture in which the thumb touched the left, middle, and right side of the forehead meant "king," while a gesture in which the hand encircled the face meant "beautiful girl." Revivals of nineteenth-century ballets still include long portions of pantomime, in which stylized gestures are used to tell the story of the ballet. Other examples are evident in special groups such as an army unit, or pilots, or other groups who communicate over long distances. In games such as charades, gestures which are consensually defined would be classified as formal pantomimic gestures, while other gestures used in charades, which are not consensually defined, fall into the category of improvisational pantomimic gestures.

2. *Improvisational pantomimic gestures.*[24] Although we have no reason to believe that these forms have as much prescribed meaning as do formal pantomimic gestures, meaning seems to accrue in a particular context as a function of *some perceptual attribute* of an object, of an action performed on or with an object, or of the reaction to an object or event. The improvisational pantomimic gestures also have a noun or event quality, but since the meaning ascribed to the gesture is dependent on the situation, context, or content, these particular gestures occur most frequently in a context which has co-occurring verbal communication. For the most part, improvisational pantomimic gestures seem to serve the function of emphasizing, concretizing, or focusing on a particular aspect of the communication occurring verbally. Examples of gestures within this category of communicative movements are drawing a number of circles in the air around one point to depict the concept concentric and swinging an imaginary bat. There is a special set of circumstances in which improvisational pantomimes seem most prominent; that is, when more typical communicative patterns are blocked—for example, when speaking to a foreigner.

B. Semantic modifying and relational gestures. These gestures usually accompany speech and are hypothesized to (*a*) serve the function of modification and specification of the communication, the same function served by adjectives and adverbs in the verbal channel; (*b*) specify the speaker's relationship to the addressee and to his communication; or (*c*) specify the relationship of one aspect of the communication to another aspect of the communication. The occurrence of these nonpantomimic gestures seems to be dependent on the communicative content co-occurring in other channels, but the forms of the gestures, when they do occur, seem to be relatively invariant and appear to have independent meaning. In general, the dimensions of these nonpantomimic gestures, along which variations seem to indicate differences in communicative significance, are (*a*) pointing and its direction, (*b*) orientation of palms, (*c*) semantic forms, and (*d*) location and size of gestures.

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[24] There is an apparent contradiction in including a set of gestures labeled improvisational under a rubric of shared code elements. Our decision to include these movements as possible communication components derives from an assumption that the particular addressor and addressee share a knowledge of the particular object or event and their attributes; objects and events referred to explicitly in the verbal content. Even though the particular movements themselves are not likely to occur again in other contexts or with other addressees who do not have the shared experience of the object or event, in the instances where they do occur, the referent (the object) is posited to be shared knowledge.

1. *Deictic movements* are pointing movements which can be done with the index finger, two or more fingers, or all fingers. Deictic movements are hypothesized to serve the function of specifying or reducing the ambiguity of the verbal referent. For example, the pronoun "you" can mean a specific person or a generalized other. When accompanied by a pointing to the addressee the meaning of "you" is the addressee; if pointed somewhere else than at the addressee it means "other than you, but not present" or "not I." Pointing gestures are also used to specify the relative positions of objects in space or events in time—"not here, not now" instances.
2. *Orientation of palms*—The position of the palms in a gesture indicates the speaker's relationship to the addressee—for example, mutual versus nonmutual—and/or about the speaker's attitude toward his communication—certain versus uncertain. Palms up is equivalent to uncertainty or to "I think" or "I believe" or "It seems to me" in a verbal statement, and adds for the addressee the message that the issue need not be pursued since noncertainty is indicated. Palms down indicates an assertion with the speaker again communicating that the subject matter is not open to question—equivalent verbal statements are "clearly," "absolutely," "without doubt," etc. Palms out (i.e., facing toward the addressee) is a statement of assertion and is equivalent to the statement, "I shall say it" or "Don't interrupt." These last gestures also seem to function as regulators of the communication interaction.
3. *Semantic forms*—Five gestures or gestural patterns have been isolated: (*a*) circling gestures, (*b*) oscillation gestures, (*c*) arhythmic "chopping" gestures, (*d*) rhythmic "chopping" gestures, and (*e*) expansion-contraction gestures. Each of these will be described briefly and its hypothesized, communicative significance posited.

   a. Circling gestures—a slow, continuous series of circular movements of any part of the hand including the arm which is hypothesized to indicate nonspecificity, globality, or generality of the verbal components in the communication. In words, this gesture is posited as indicating "I mean more than the specific words I have used," or "The specific referent term is only one general attribute of the event I'm describing."

   b. Oscillation gestures—a series of slow to moderate, semicircular, back and forth movements of the hand which seem to indicate "either or" or "one or the other" of two components, or "on one hand versus on the other hand."

   c. Arhythmic chopping gestures—linear, staccatolike movements of the arm and hand, usually performed in a limited series of one or two in which the same plane is maintained throughout the movement. The chopping gestures seem to be associated with the speaker's experience of discreteness of the elements or contents of his verbal communication. For example, a verbal statement in the form of "We have to consider A and B and C" is likely to be accompanied by a series of three "chops"—one for each component. The verbal statement in the form, "We have to consider John and his behavior" with a single chop at the end is interpreted to mean only his behavior—discreteness indicator—versus two chops where John in general is to be considered and his behavior is to be considered separately.

   d. Rhythmic chopping gestures—also linear, staccato movements, usually *more* than one or two, which have a rhythm independent of the rhythm of the co-occurring verbal content. In this form, the staccato chop is interpreted to mean emphasis rather than discreteness.

   e. Expansion-contraction gestures—a slow expansion and contraction of the arm in which the addressor looks as if he were playing a concertina. Typically these movements indicate a size or extent dimension of the referent carried in the verbal channel.

   In sum, the semantic forms seem to serve the function of specification and modification served by adjectives and adverbs in the verbal channel of communication; the palms seem to indicate the speaker's relationship to his communication, the addressee, or the relationship of one part of his communication to another part of his communication. However, much of the information communicated through the gestural channel appears to be redundant with that communicated through other channels. For example, tonal changes are effective indicators of the emphasis a speaker wishes to place on any portion of his verbal communication; facial expressions can communicate doubt or uncertainty as easily as gestures. The chief communicative functions of the gestural channel appear to be to specify the referent of an ambiguous verbal statement, to specify the addressor's relationship to his verbal communication, and to indicate intensity or emphasis by introducing redundancy into the message.

It is further hypothesized that the size and location of the gesture are co-varying attributes which indicate the speaker's experience of the centrality or the importance of his gestural communication. A small gesture made in the plane directly in front of the speaker is taken as equivalent in communicative significance to a larger gesture made more peripherally, the view being that a gesture must be seen to serve a communicative function. In line with this hypothesis, it would seem that the speaker's experience of the centrality or importance of the communicative component carried gesturally will determine both the size and location of the gesture. From this latter hypothesis comes the derived statement that variations in the size and location of the gesture are associated with variations in the emphasis or intensity placed on the gestural component of the message.

Preliminary research has been concluded using this set of descriptive categories. In one study, untrained subjects were shown videotaped exemplars of each of the gestures and were asked what they took each gesture to mean; the subjects, as predicted, assigned to the gestures the appropriate meaning category (Rubinow, 1970). Interestingly, there were some intriguing individual differences in the reliability of these categorizations. In a second study using videotape recordings of interviews without the sound on, trained judges were able to discriminate the gestures from one another with a high degree of reliability (Novick, 1967).

Up to this point we have (*a*) specified a set of events (search, correction, regulators, and message modulations) whose occurrence can be taken as independent evidence of code and code usage, (*b*) elaborated a set of criteria for selecting behaviors as possible components in a nonverbal communication code, and (*c*) described some hand and arm movements as exemplars of the kinds of behaviors which might be investigated in nonverbal encoding. What remains to be done, then, is to outline a possible research strategy for determining whether the posited nonverbal code components occur predictably as a function of changes in the experiences to be made public, of the conditions of the communication exchange, of the addressee regulators, or of the subsequent verbal behaviors of the addressee when the addressor's use of verbal forms is restricted. In addition to demonstrating the predictability of the occurrence of the nonverbal forms, it is necessary to demonstrate that the addressee's decoding response to the occurrence of the nonverbal forms is equivalent to his decoding response to a verbal utterance where use of the verbal forms is not restricted.

Below is an outline of some possible manipulations which could be attempted within our proposed strategy.

A. Manipulation of the experience to be made public.
   1. A subject is required to talk about a subject matter he knows well and about a subject matter he knows less well (e.g., a psychologist is required to talk about some subject in psychology and about some subject in physics). We would predict an increase in the number of uncertainty gestures (i.e., palms up) and of vagueness or generality gestures (i.e., circling) when the subject talks about less familiar subject matter.
   2. A subject is required to say statements with which he agrees and statements with which he disagrees. We would expect more rhythmic chops and palms down in the former case than in the latter case. Incidentally, we would also expect that the gestures which co-occur with agreement statements would be larger and/or would be located in a more central plane.

B. Manipulation of the conditions of communication.
   1. The number of channels available by which a subject can communicate could be varied, as when, for example, a subject is required to speak in a foreign language he has just begun to learn. We assume far less verbal fluency for the second language than for a first language and would predict increased numbers of pantomimes, palms up, and circling gestures for the second language condition.
   2. A subject is to communicate face-to-face and via telephone to the same addressee. We predict more semantic modifiers and pantomimes in the face-to-face condition than for the telephone condition, with a concomitant decrease in verbal explicitness for the face-to-face exchange.

C. Manipulation of addressee variables.
   1. A subject would speak about the same things with two addressees, one of whom did not emit the customary decoding regulators. We predict that in the absence of

these regulators, the addressor will begin to use more gestures, particularly more pantomimes and more palms up, and will change the verbal components radically.

2. A subject is required to speak to an adult who is identified as a native speaker and an adult who is identified as a foreigner. We predict a marked increase in pantomimic gestures for an addressee who is assumed to have difficulty in decoding. We would predict similar changes for addressees who are children.

The number of possible experimental manipulations and testable hypotheses is dependent only on the investigator's ingenuity. Further, as we have noted earlier, we hold that this strategy should be applicable to the study of any channel in communication, with the stipulations that addressors and addressees be drawn from the same communication group and that each addressor be assigned to two conditions (e.g., restriction on verbal code use vs. no restriction). The responses of the addressees for the two conditions can then be compared for decoding equivalence.

Once code and code usage can be specified for a given channel, it should be possible to investigate (*a*) developmental differences in code usage (e.g., deictive movements and informal pantomimic gestures are expected to occur earlier than the semantic modifying and relational gestures); (*b*) individual differences in code usage (e.g., verbal articulateness and amount of gesture are expected to be inversely related); (*c*) subgroup differences in code usage and responses, as well as in the use of and response to regulators (e.g., Brooks, Brandt, & Wiener, 1969, have found such subgroup differences in response to tonal inflection); and (*d*) differences in code usage for groups with differential psychiatric diagnoses (e.g., obsessive-compulsives are expected to use more gestures of enumeration and relationships; schizophrenics are expected, as nonconsensual communicators, to use few, if any, gestures and to respond to—that is, decode—few gestures).

In sum, our proposed approach attempts to emphasize encoding in the study of communication rather than relying on a decoding perspective, and attempts to operationalize the concepts of encoding, decoding, code, and communication. This attempt will, we hope, make possible more systematic study of nonverbal communication than has generally occurred. It is also hoped that the making explicit of one alternative approach will stimulate others to offer other alternative approaches to the study of nonverbal communication.

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